Question	Answer	Marks	Partial Marks
12(b)	15.6 or 15.55 to 15.56 nfww	**3**	**M2** for 70 ÷ (*their* final time in hours) (final time =) $1.5 + 0.5 + \frac{70 - their\,30}{16}$ or 4.5 or *their* final time from graph or **M1** for 70 ÷ any time
13	$\frac{33}{8}$ or $\frac{17}{6}$ $\quad 2\frac{1}{8} - \frac{5}{6}$	**B1**	Correct step for dealing with mixed numbers Allow $\frac{33k}{8k}$ or $\frac{17k}{6k}$
	$\frac{99}{24}$ and $\frac{68}{24}$ $\quad [2]\frac{3}{24} - \frac{20}{24}$	**M1**	Correct method to find common denominator e.g. $4\frac{3}{24}$ and $2\frac{20}{24}$
	$1\frac{7}{24}$ cao and correct working	**A1**	
14	2.6[0] or 2.600…	**3**	**M2** for $\sqrt[10]{\frac{1328.54 + 4540}{4540}}$ or **M1** for $4540 \times k^{10} = 1328.54 + 4540$ for any k If 0 scored **SC1** for answer –11.6 or –11.56…
15	$4a^2b$ final answer	**2**	**M1** for two correct parts out of three from 4, a^2 and b in final answer
16(a)	$(M \cup G) \cap P'$	**1**	
16(b)	22	**1**	
16(c)	$\frac{8}{23}$ oe	**2**	**M1** for $\frac{k}{23}$ or $\frac{k}{3+9+5+6}$ or $\frac{8}{c}$ or $\frac{3+5}{c}$ $c \neq 1$ or for 8 and 23 identified
17(a)	 Correct sketch to go through (0, 0), (180, 0) and (360, 0)	**2**	**B1** for correct sine curve shape through the origin

Question	Answer	Marks	Partial Marks
17(b)	199.5 or 199.47… and 340.5 or 340.52 to 340.53…	**3**	**B2** for one correct or **M1** for $\sin x = -\frac{1}{3}$ oe If 0 scored **SC1** for two reflex angles with sum of 540 or two non-reflex angles with sum of 180
18(a)	2.5	**3**	**M1** for $y = k \times \sqrt[3]{x+1}$ **M1** for $y = their k \times \sqrt[3]{124+1}$
18(b)	multiplied by 4 oe	**1**	
19(a)	$\frac{x+8}{7}$ final answer	**2**	**M1** for $x = 7y - 8$ or $y + 8 = 7x$ or $\frac{y}{7} = x - \frac{8}{7}$
19(b)	4	**2**	**M1** for $4 \div \frac{1}{3} + 5$ oe or better
20(a)	$(2m+3p)(1-4k)$ final answer	**2**	**B1** for $2m + 3p - 4k(2m+3p)$ or better or $2m(1-4k) + 3p(1-4k)$ or correct answer seen and spoilt
20(b)	$5(x-2y)(x+2y)$ final answer	**3**	**B2** for $(5x - 10y)(x + 2y)$ or $(x - 2y)(5x + 10y)$ or correct answer seen then spoilt or **B1** for $5(x^2 - 4y^2)$ or for $(x-2y)(x+2y)$
21	[a =] 2 [b =] − 1	**5**	**M2** for correct method to find two simultaneous equations e.g. two from $a \times 1^2 + b \times 1 - 4 = -3$ $a \times 2^2 + b \times 2 - 4 = 2$ $3a + b = 2 - -3$ or **M1** for 1 correct equation **M1** for correctly eliminating one variable for *their* simultaneous equations **A1** for a = 2 **A1** for b = − 1
22	4 : 3 oe	**2**	**M1** for $\overrightarrow{AD} = -\frac{4}{7}x + \frac{4}{7}y$ oe or $\overrightarrow{DB} = -\frac{3}{7}x + \frac{3}{7}y$ oe